Soham Sarkar

Ph.D. Candidate in Applied & Computational Mathematics
Department of Mathematical Sciences · The University of Texas at Dallas
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Research Interests

Applied and Computational Mathematics: nonlinear wave propagation and dissipative systems; mode-locked and ultrafast fiber lasers; semiconductor saturable absorber mirrors (SESAMs); optical frequency combs; spectral and linear stability analysis; numerical continuation and bifurcation; numerical PDEs (Fourier split-step, finite-difference, finite-element); adjoint-state optimization; stochastic modeling of laser noise and timing jitter; high-performance computing; dynamical systems.

Research Summary: My dissertation develops computational and spectral methods for analyzing the stability and noise sensitivity of periodically stationary pulses in lumped models of passively mode-locked fiber lasers with slow saturable absorbers. Starting from Haus-type nonlinear PDE models, I compute essential and discrete spectra of linearized operators, use numerical continuation to map stability boundaries and bifurcations, and apply adjoint-state quasi-Newton (BFGS) optimization to locate optimal solutions. I am extending this framework to stochastic models with additive noise to quantify timing jitter and power spectral density, with the goal of supporting the search for low-noise frequency-comb sources.

Education

Ph.D. in Applied & Computational Mathematics Expected May 2027
The University of Texas at Dallas, Richardson, TX, USA
Dissertation Title: Spectral Stability of Breathers in Ultrafast Lasers with Slow Saturable Absorber
Co-Advisors: Dr. John Zweck & Dr. Dmitry Rachinskiy
Oral Research Capability Exam (Dissertation Proposal): Passed May 2024
Qualifying Examination: Passed August 2023

M.S. in Applied Mathematics August 2025
The University of Texas at Dallas, Richardson, TX, USA

M.Sc. in Mathematics April 2020
Savitribai Phule Pune University, Pune, Maharashtra, India
Specialization: Commutative Algebra & Algebraic Graph Theory
Master's Thesis: Molien's Theorem to Compute Invariants

B.Sc. (Honours) in Mathematics June 2018
University of Calcutta, Kolkata, West Bengal, India
Minors: Physics and Chemistry

Publications & Preprints

Manuscripts in Preparation

1. **S. Sarkar**, J. Zweck, and D. Rachinskiy, "Spectral methods to analyze noise in ultrafast lasers," *In Preparation.*
2. **S. Sarkar** and J. Zweck, "Spectral stability in lumped laser model with slow saturable absorber," *In Preparation.*

Master's Theses and Monographs

3. **S. Sarkar**, "Molien's Theorem to Compute Invariants on Graded Algebras," *Master's Monograph*, Savitribai Phule Pune University, 2020.
4. **S. Sarkar**, "Expander Graph and its Applications," *Master's Monograph*, Savitribai Phule Pune University, 2020.

Research Experience

Doctoral Researcher & Graduate Research Assistant Aug 2023 - Present
Department of Mathematical Sciences, UT Dallas Richardson, TX
Advisors: Dr. John Zweck (New York Institute of Technology) and Dr. Dmitry Rachinskiy (UT Dallas)

- Developed MATLAB/Simulink computational models for lumped ultrafast fiber laser systems incorporating fiber amplifiers, single-mode and dispersion-compensating fibers, output couplers, and Semiconductor Saturable Absorber Mirrors (SESAMs) governed by Haus-type nonlinear PDEs.
- Modeled two-level energy dynamics of slow saturable absorbers and contrasted pulse behavior against fast absorber models to quantify response-time effects on laser pulse stability.
- Implemented high-order numerical PDE solvers using Fourier split-step, finite-difference, and finite-element schemes to simulate pulse propagation across experimentally grounded physical parameters.
- Performed linear and spectral stability analysis by computing essential and discrete spectra of linearized operators, employing numerical parameter continuation to identify stability boundaries and bifurcations.
- Formulated analytical adjoint equations and implemented quasi-Newton (BFGS) optimization algorithms to accelerate convergence to stationary periodic pulse solutions, reducing computational cost compared to explicit operator discretization.
- Investigated stochastic laser dynamics incorporating additive white Gaussian noise to analyze timing jitter and power spectral density using Monte Carlo simulations and eigenfunction expansions.
- Investigated passive mode-locking pulse stability under NSF-funded research (NSF DMS-2106203, PI: Dr. John Zweck; Spring 2024) to evaluate low-noise optical frequency comb generation.
- Deployed parallelized large-scale PDE simulation suites on the UT Dallas `Ganymede2` HPC cluster using OpenMPI and SLURM.

Postgraduate Research Fellow Feb 2021 - Aug 2021
Savitribai Phule Pune University & IISER Pune Pune, India

- Investigated zero-divisor graphs of commutative rings and partially ordered sets under Prof. Vinayak Joshi.
- Constructed line graph characterizations of subgroup lattices of cyclic groups.
- Defined graph isomorphisms between the comaximal subgroup graph lattice of a group and the zero-divisor graph of its dual lattice.

Talks & Presentations

Conference Talks and Posters

1. **Talk & Poster Presentation:** *"Stability of pulses in a lumped model of a laser with a slow saturable absorber,"* Midwest Numerical Analysis Day, University of Notre Dame, IN April 2026
2. **Poster Presentation:** *"Modeling Stability in Nonlinear Optics,"* SIAM TX-LA Sectional Meeting, UT Austin, TX Sept 2025
3. **Contributed Lightning Talk:** *"Stability in a Short Pulse Laser,"* Dynamical Systems and Fractal Geometry Conference, University of North Texas, TX May 2024

Seminars and Departmental Presentations

4. **Graduate Student Seminar:** *"Weekly Partial Differential Equations Meet,"* covering *PDE I: Basic Theory* (Taylor), Department of Mathematical Sciences, UT Dallas, TX Fall 2025 - Present
5. **Course Presentation:** *"Heart Failure Prediction (using Linear & Logistic Regression Models),"* Department of Mathematical Sciences, UT Dallas, TX April 2025
6. **Dissertation Proposal Presentation (ORCE):** *"Stability of Short Pulse Lasers with Realistic Saturable Loss and Amplification,"* Advisory Committee Examination, UT Dallas, TX May 2024
7. **Seminar Presentation:** *"Probability and Information Theory for Machine Learning,"* Weekly Reading Group on *Deep Learning* (Goodfellow et al.), Department of Mathematical Sciences, UT Dallas Fall 2023
8. **Seminar Talk:** *"Mallat's Theory on Multiresolution Signal Decomposition and Wavelets,"* Image & Signal Analysis Journal Club, Department of Mathematical Sciences, UT Dallas, TX Fall 2022

Teaching Experience

Graduate Teaching Assistant Aug 2022 - Present
Department of Mathematical Sciences, UT Dallas Richardson, TX

- Supported and mentored more than 400 undergraduate and graduate students across 13+ course sections (12 distinct courses).
- Conducted active learning problem sessions built on board work, think-pair-share, and small-group discussion; designed homework assignments, created quizzes, and held regular office hours.
- **Primary Teaching Assistant / Recitation Leader:**
  - Differential Equations with Applications (`MATH 2420`; incl. Collegium V Honors) Fall 2026, 2024
  - Linear Algebra (`MATH 2418`) Summer 2024
  - Calculus of Several Variables, Active Learning Model (`MATH 2415`) Fall 2023
  - Calculus I (`MATH 2417`) Fall 2025, Summer 2023
  - Integral Calculus (`MATH 2414`) & Differential Calculus (`MATH 2413`) Spring 2023, Fall 2022
- **Teaching Assistant / Grader for Advanced Undergraduate & Graduate Courses:**
  - Elementary Analysis II (`MATH 5302`) & Mathematical Analysis II (`MATH 4302`) Spring 2025
  - Mathematical Analysis I (`MATH 4301`) Summer 2025
  - Numerical Analysis (`MATH 4334`) Fall 2026, 2025, 2024; Spring 2026, 2025
  - Applied Calculus I & II (`MATH 1325`, `1326`) Fall 2022, Spring 2023
- **Pedagogical Innovation:** Designed computational laboratory demonstration modules in MATLAB illustrating numerical algorithms, stability analysis, and scientific visualization for applied mathematics students.
- **Teaching Interests:** calculus, linear algebra, differential equations, real analysis, numerical analysis, scientific computing, optimization, and dynamical systems; a topics course on mathematical modeling and stability analysis in nonlinear optics.

Subject Matter Expert in Advanced Mathematics Feb 2021 - July 2022
Chegg Inc. Remote

- Authored over 500 step-by-step analytical and numerical solutions for advanced university-level mathematics, differential equations, and computational methods problems.

Honors, Awards & Funding

Fellowships and Funding

- **NSM Summer Research Fellowship**, UT Dallas 2026
- **PVST Doctoral Tuition Scholarship**, Natural Sciences & Mathematics, UT Dallas 2022 - 2027
- **Assistantship supported by NSF Award DMS-2106203** (PI: J. Zweck), UT Dallas 2024
- **Postgraduate Research Fellowship**, SPPU & IISER Pune 2021

Travel Awards: Midwest Numerical Analysis Day, University of Notre Dame (2026); SUMIRFAS, Texas A&M University (2025); Dynamical Systems & Fractal Geometry Conference, UNT Denton (2024); Texas Analysis and Mathematical Physics Symposium (TexAMP), Texas A&M University (2024).

National Examinations (India)

- All India Ranks **4**, **6**, and **9** in the IOMA, SPPU, and TUEE M.Sc. entrance examinations (2018) and Rank **5** in the DUET Ph.D. entrance examination (2020); qualified GATE (2022) and JAM (2018).

Technical & Computational Skills

- **Mathematical & Laser Modeling:** Ultrafast Fiber Lasers, Nonlinear Wave Propagation, Semiconductor Saturable Absorbers (SESAM), Ginzburg-Landau & Haus Equations, Energy Level Dynamics.
- **Numerical & Computational Methods:** Numerical PDEs, Fourier Split-Step Solvers, Spectral & Finite Difference Algorithms, FEM, Linear Stability & Eigenvalue Analysis, Parameter Continuation, Adjoint

Optimization, Monte Carlo Simulation.

- **Programming Languages:** MATLAB, Simulink, Python (NumPy, SciPy, Pandas, PyTorch, Keras), C, C++, R (ggplot2, dplyr, tidyr), SQL, HTML/CSS, Scala.
- **HPC & Typesetting Software:** MPI/OpenMPI, SLURM, Linux/UNIX, Docker, Git, LaTeX, Jupyter, Ganymede2 & Juno (UT Dallas clusters), Google Colab.

Professional Service & Memberships

Communication & Outreach Coordinator Sept 2024 - Sept 2025
SIAM UTD-SMU Student Chapter (Society for Industrial and Applied Mathematics) UT Dallas

- Coordinated computational mathematics seminars, career panels, and invited industry talks between UT Dallas, Southern Methodist University (SMU), and external researchers.

Professional Membership: SIAM, student member Aug 2022 - Present

Conferences & Workshops Attended

- **Midwest Numerical Analysis Day**, University of Notre Dame, IN April, 2026
- **Complex (Analysis & Dynamics II)**, UT Dallas, TX March, 2026
- **Complex (Analysis & Dynamics I**, UT Dallas, TX Oct, 2025
- **SIAM TX-LA Sectional Meeting**, UT Austin, TX Sept, 2025
- **Workshop in Analysis & Probability**, Texas A&M University, TX July - Aug, 2025
- **Dynamical Systems and Fractal Geometry Conference**, UNT Denton, TX May 2024
- **Texas Analysis and Mathematical Physics Symposium**, Texas A&M University, TX Feb 2024
- **IIT Bombay Virtual Commutative Algebra Seminar Series**, IIT Bombay, India April 2022
- **Facets of CACAAG (Combinatorial Aspects of Commutative Algebra)**, IIT Bombay, India 2022
- **Annual Foundation School (AFS-II & AFS-I)**, National Center for Mathematics (NCM), India 2021 & 2019
- **Cyclic Cohomology at 40**, Fields Institute for Research in Mathematical Sciences, Canada Sept 2021
- **International Conference on Discrete Groups, Geometry and Arithmetic**, TIFR Mumbai, India Aug 2021
- **Introduction to Algebraic Analysis**, Osaka University International Summer Program July 2021
- **MOSS (Mathematics Online Seminar Series)**, IISER Berhampur, India Aug 2020
- **Workshop on Analysis and Applications**, Jointly Organized SPPU, IISc Bangalore, IIT Gandhinagar & UM-DAE CEBS, India Nov 2019

Selected Graduate Coursework (UT Dallas)

Analysis & Applied Mathematics: Real Analysis (`MATH 6301`); Functional Analysis I (`6302`); Theory of Complex Functions I (`6303`); Principles & Techniques in Applied Mathematics (`6319`).
Differential Equations & Dynamical Systems: Ordinary Differential Equations (`MATH 6315`); Differential Equations (`6316`); Partial Differential Equations I (`7313`); Applied Dynamical Systems I (`6324`).
Numerical Methods & Scientific Computing: Numerical Analysis (`MATH 6313`); Numerical Linear Algebra (`6340`); Numerical Analysis of Differential Equations (`6318`); Scientific Computing (`6342`); Optimization (`6321`).
Statistics & Data Science: Statistical & Machine Learning (`STAT 6340`); Probability & Statistics for Data Scientists & Bioinformatics (`STAT 5353`); Introduction to Big Data Management (`CS 6307`).

Certifications

- **Using Python to Access Web Data**, University of Michigan (Coursera) June 2021
- **Python Data Structures**, University of Michigan (Coursera) June 2021
- **Programming for Everybody**, University of Michigan (Coursera) May 2021